

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0711

Parkin 138 kV
April 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0711. The Interconnection Customer proposes a 92.8 MW solar photovoltaic facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on December 04, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 294 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,604,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CARMEL 146TH ST – OAK CRK 138kV	CLOVER – BEAR 345kV	153.6	200	6.5
CEIBA BAJA 7012 – CARMEL 146TH ST 138kV	BEAR – CLOVER 345kV L	152.2	204	6.5
NEW LONDON – BARKSDALE 138kV	S CRK – WYTHE 138kV	142.1	150	5.3
TANGY – CEIBA BAJA 7012 138kV Line	BEAR – CLOVER 345kV	140.2	229	6.5
ALDERWOOD – CARON ICE COGEN 138kV	S CRK – WYTHE 138kV	111.3	271	5.6
S CRK – WYTHE 138kV	PARKIN – CANDLERS MTN 138kV Line	110.1	517	46.1
RED HILL – GRASSMERE 138kV line	BEAR – CLOVER 345kV	108.9	295	6.7
COAL MTN – GRASSMERE 138kV	BEAR – CLOVER 345kV	106.6	329	6.7
COAL MTN – CENTRAL AUSTIN 138kV	BEAR – CLOVER SUBSTATION 345kV	106.1	338	6.7
CENTRAL AUSTIN – WAYNESBORO 138kV Line	CLOVER – BEAR 345kV	105.5	348	6.7
WAYNESBORO – ALDERWOOD 138kV	CLOVER – BEAR 345kV L	104.1	377	6.7
BOONSBORO – AMARGO 138kV Line	OPOSSUM CREEK SUBSTATION – PARKIN 138kV	103.1	723	100.0
BROOKVILLE SUBSTATION – AMARGO 138kV	PARKIN – OPOSSUM CREEK 138kV Line	102.6	747	100.0
UNITY SUBSTATION – WYTHE 138kV	WYTHE – S CRK 138kV	102.5	244	44.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SOUTH LYNCHBURG – BROOKVILLE 138kV line	PARKIN – OPOSSUM CRK 138kV	101.9	785	100.0
SOUTH LYNCHBURG – CANDLERS MTN 138kV	OPOSSUM CREEK – PARKIN 138kV Line	101.7	797	100.0
CENTRAL TIE – WYTHE 138kV	CANDLERS MTN – PARKIN 138kV	101.6	798	100.0
CENTRAL TIE – OPOSSUM CREEK 138kV line	PARKIN – CANDLERS MTN 138kV	101.4	809	100.0
CANDLERS MTN – PARKIN 138kV	PARKIN – OPOSSUM CREEK 138kV	101.2	821	100.0
OPOSSUM CREEK – PARKIN 138kV	PARKIN – CANDLES MOUNTAIN 138kV Line	101.2	821	100.0
LEESVILLE – CLOVER 345kV	PERSON – STATE SCHOOL 345kV	100.6	1,241	14.0
INTEL – AIBONITO PDS SUBSTATION 138kV	S CRK – AIBONITO PS 138kV	100.4	277	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARMEL 146TH ST – OAK CRK 138kV	Rebuild line with higher-capacity conductor	\$13,422,000	8% (\$1,040,000)
CEIBA BAJA 7012 – CARMEL 146TH ST 138kV	Rebuild line with higher-capacity conductor	\$10,963,000	8% (\$849,000)
NEW LONDON – BARKSDALE 138kV Line	Rebuild line with higher-capacity conductor	\$6,191,000	7% (\$444,000)
TANGY – CEIBA BAJA 7012 138kV Line	Rebuild line with higher-capacity conductor	\$4,879,000	8% (\$378,000)
ALDERWOOD – CARON ICE COGEN 138kV	Rebuild line with higher-capacity conductor	\$7,228,000	7% (\$509,000)
S CRK – WYTHE 138kV	Rebuild line with higher-capacity conductor	\$11,907,000	10% (\$1,133,000)
GRASSMERE – RED HILL SUBSTATION 138kV	Reconductor line	\$996,000	6% (\$65,000)
COAL MTN – GRASSMERE 138kV	Reconductor line	\$1,398,000	6% (\$91,000)
COAL MTN – CENTRAL AUSTIN 138kV	Reconductor line	\$2,328,000	6% (\$151,000)
CENTRAL AUSTIN – WAYNESBORO 138kV Line	Reconductor line	\$982,000	6% (\$64,000)
ALDERWOOD – WAYNESBORO SUBSTATION 138kV	Reconductor line	\$1,688,000	6% (\$110,000)
BOONSBORO – AMARGO 138kV	Reconductor line	\$1,867,000	11% (\$209,000)
BROOKVILLE SUBSTATION – AMARGO 138kV	Reconductor line	\$314,000	11% (\$35,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WYTHE SUBSTATION – UNITY 138kV	Reconductor line	\$2,846,000	10% (\$288,000)
SOUTH LYNCHBURG – BROOKVILLE 138kV line	Reconductor line	\$1,396,000	11% (\$157,000)
CANDLERS MOUNTAIN – S LYNCHBURG 138kV	Reconductor line	\$1,159,000	11% (\$130,000)
CENTRAL TIE – WYTHE 138kV	Reconductor line	\$1,238,000	11% (\$139,000)
CENTRAL TIE – OPOSSUM CREEK 138kV line	Reconductor line	\$557,000	11% (\$62,000)
PARKIN – CANDLERS MTN 138kV	Reconductor line	\$567,000	11% (\$64,000)
OPOSSUM CREEK – PARKIN 138kV	Reconductor line	\$500,000	11% (\$56,000)
LEESVILLE SUBSTATION – CLOVER 345kV	Reconductor line	\$69,488,000	2% (\$1,329,000)
INTEL – AIBONITO PDS 138kV	Reconductor line	\$3,573,000	8% (\$301,000)
	Total allocated to project		\$7,604,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.